Chapter 78 — Order Tracking & Live Status

> **Part IX — Feature Architectures** · Complexity: **Standard** (≥25 interview questions)
>
> **Prerequisites:** Ch 49 (FCM/push), Ch 63 (WebSocket), Ch 32 (sync), Ch 3 (SSOT), Ch 67 (live location), Ch 50 (notifications), Ch 44 (lifecycle). Pairs with Ch 75 (checkout), Ch 66 (SSE).
>
> **Reference surface in StreamCart:** after checkout (Ch 75) — the order-status screen and live delivery tracking ("Confirmed → Packed → Shipped → Out for delivery → Delivered", plus a moving driver on a map) — a **server-driven status machine** kept fresh via push, sync, and (while watching) a live channel, with notifications.

78.1 Problem Definition and Real-World Context

After an order is placed (Ch 75), the user wants to **watch its progress**: a status timeline (confirmed, packed, shipped, out-for-delivery, delivered) and, for delivery, a **live driver location** on a map. The architecture is a composition of the book's real-time and sync layers applied to a **server-owned status state machine**: the status is **authoritative on the server**; the client **reflects** it, kept fresh by the right **delivery mechanism per app state** — **push/FCM** (Ch 49) for background updates + notifications, **sync** (Ch 32) as the reliable backstop, and a **live channel** (WebSocket Ch 63 / SSE Ch 66) only **while the user is actively watching** for low-latency updates and the moving driver (Ch 67).

Concretely, teams hit:

- **Polling for status.** Polling the order endpoint is laggy and wasteful; status updates should **push** (FCM, Ch 49) and the client should **sync** (Ch 32), with a **live channel** only while watching.
- **Client-invented status.** Inferring/advancing status on the client is wrong — status is a **server-owned state machine** (valid transitions, timestamps); the client **renders** it, never invents it.
- **Wrong mechanism per state.** Holding a WebSocket open in the background drains battery (Ch 63/47); not using a live channel while watching makes the driver map laggy. Match the mechanism to **app state**: live channel (foreground/watching) + FCM (background) + sync (reconcile).
- **Missed/stale updates.** A dropped push or missed live event must **self-heal** via sync (Ch 32) — the order SSOT (Ch 3) is reconciled against the server; status is never lost.
- **Notifications.** Status changes drive **notifications** (Ch 50) — "Your order shipped" — that must be consistent with the in-app status (one source), and work when backgrounded (FCM, Ch 49).
- **Live location (delivery).** The moving driver is **live location** (Ch 67) streamed server→client **only during active delivery while watching**, with smoothing and a map.
- **Consistency & history.** The order status + ETA + history must be **consistent** (SSOT), survive process death, and show accurately across devices.

The architecture: model order status as a **server-owned state machine** reflected in the **SSOT** (Ch 3); keep it fresh with the **right mechanism per app state** — a **live channel** (WebSocket Ch 63 / SSE Ch 66) **only while actively watching** (low-latency status + driver map, Ch 67), **FCM** (Ch 49) for **background** status changes + notifications (Ch 50), and **sync** (Ch 32) as the **reliable backstop** so dropped updates self-heal; render the **status timeline + ETA**

+ live map; and keep it **consistent, death-safe, and notification-driven**. This chapter composes the real-time and sync stack onto a server-driven order lifecycle.

Three practical use cases

1. **Order status timeline (StreamCart).** Confirmed → Packed → Shipped → Out-for-delivery → Delivered, server-owned, reflected in the SSOT; FCM + notification on each change; sync reconciles.
2. **Live delivery map.** While the user watches the "out for delivery" screen, a live channel (Ch 63/66) streams the driver's smoothed location (Ch 67) and ETA; closes when they leave.
3. **Background + reopen.** The app is closed when the order ships; FCM delivers the status + notification (Ch 49/50); on reopen, sync shows the current state (Ch 32).

Why this is architectural

Order tracking is a **clean composition** of a server-owned status machine (Ch 3) with the real-time/sync mechanism-selection discipline: FCM (Ch 49) + live channel (Ch 63/66) + sync (Ch 32) + live location (Ch 67) + notifications (Ch 50), each used per app state. The decisions — server-owned status, mechanism-per-state, sync-as-backstop, watching-scoped live channel — determine freshness, battery, and correctness.

78.2 Requirements Engineering

Functional requirements

ID	Requirement
F1	Reflect a server-owned order status state machine in the SSOT; never invent status (Ch 3).
F2	Keep status fresh per app state: live channel (watching) + FCM (background) + sync (backstop) (Ch 63/49/32).
F3	Drive notifications on status changes, consistent with in-app status (Ch 49/50).
F4	Stream live driver location + ETA while actively watching (Ch 67).
F5	Self-heal missed/dropped updates via sync; survive process death (Ch 32/8).
F6	Consistent across devices; accurate timeline/ETA/history.

Non-functional requirements

Attribute	Policy
Freshness	Low-latency while watching; prompt when backgrounded (push).
Battery	Live channel only while watching; FCM for background (Ch 63/47/49).
Correctness	Server-owned status; sync backstop; no lost updates.
Consistency	One SSOT; notification matches in-app; cross-device (Ch 3/32).
Resilience	Survives process death; dropped updates self-heal.

Scope

- **In scope:** server-owned order status state machine reflected in SSOT (Ch 3), mechanism-per-state (live channel Ch 63/66 while watching + FCM Ch 49 background + sync Ch 32 backstop), status-change notifications (Ch 50), live driver location + ETA (Ch 67, watching-scoped), self-healing via sync (Ch 32), process death (Ch 8), cross-device consistency, timeline/history rendering.
- **Out of scope:** WebSocket/SSE internals (Ch 63/66), FCM internals (Ch 49), sync algorithm (Ch 32), live-location mechanics (Ch 67), notifications detail (Ch 50), checkout (Ch 75).

Assumptions & constraints

- The **server owns order status** (state machine + timestamps) and emits changes via FCM (Ch 49) + a live channel (Ch 63/66); the client reflects it in the **SSOT** (Ch 3). Live channel + driver location only **while watching** (Ch 44/67). `minSdk 24`.

78.3 System Design Fundamentals

- **Order status is a server-owned state machine (the client reflects it, Ch 3).** The order moves through **defined states** (Placed → Confirmed → Packed → Shipped → Out-for-delivery → Delivered, plus Cancelled/Returned) with **valid transitions** and **timestamps**, all **authoritative on the server**. The client **renders** the current state + timeline from the **SSOT** (Room, Ch 3); it **never advances or infers** status itself (that's the server's job). This mirrors push-as-trigger (Ch 49) and socket-as-fast-path (Ch 63): the **server is truth**.
- **Mechanism per app state (the core decision, Ch 49/63/32):**
  - **Live channel (WebSocket Ch 63 / SSE Ch 66) — only while actively watching.** When the user is **on the tracking screen**, open a live channel for **low-latency** status updates and the **moving driver** (Ch 67); **close it** when they leave (battery, Ch 44/ 47). SSE (Ch 66) suffices for one-way status; WebSocket if bidirectional.
  - **FCM (Ch 49) — background status changes + notifications.** When the app is **backgrounded/closed**, status changes arrive via **high-priority FCM** (Ch 49), updating the SSOT (via a sync) and posting a **notification** (Ch 50) — "Your order shipped."
  - **Sync (Ch 32) — the reliable backstop.** On app open/foreground and periodically, **sync** the order from the server into the SSOT so **any missed push or live event self-heals**. The order status is never lost because sync reconciles against the authoritative server state. This is the same mechanism-selection discipline as Ch 63/66 applied to orders: **live channel (watching) + FCM (background) + sync (backstop)**.
- **Self-healing (Ch 32).** Because FCM and live channels are **best-effort**, a dropped status update **self-heals**: the next **sync** (on open/periodic) pulls the authoritative status into the SSOT. Push/live are **accelerators**; **sync is the guarantee** (the order SSOT always converges to server truth). (Same principle as Ch 49/63.)
- **Notifications consistent with in-app status (Ch 50/49).** Status-change notifications are driven by the **same server status** and the **same SSOT** — tapping a "shipped" notification deep-links (Ch 52) to the tracking screen showing the matching status. One source of status → notification and screen never disagree. Notifications work backgrounded (FCM, Ch 49).
- **Live driver location + ETA (Ch 67), watching-scoped.** For active delivery, the **driver's live location** (Ch 67) streams server→client over the live channel **only while the user watches** the map; it's **smoothed** (Ch 67)

and paired with a server-computed **ETA**. Closing the screen closes the stream (battery). The driver side is a `location` FGS (Ch 67); the customer side is a watching-scoped subscriber.

- **SSOT, process death, cross-device (Ch 3/8/32).** Order status + timeline + ETA live in the **SSOT** (Ch 3) so the screen renders instantly, survives **process death** (Ch 8), and is **consistent across devices** (server-authoritative + sync, Ch 32). The timeline (with timestamps) is history the server owns and the client displays.
- **Lifecycle (Ch 44).** Open the live channel on the tracking screen's `onStart`, close on `onStop`; rely on FCM when backgrounded; sync on foreground. Never hold the live channel or driver stream in the background.

Core insight: **order tracking reflects a server-owned status state machine in the SSOT (Ch 3) and keeps it fresh with the right mechanism per app state — a live channel (WebSocket Ch 63 / SSE Ch 66) only while the user is actively watching (low-latency status + the smoothed driver location and ETA, Ch 67), high-priority FCM (Ch 49) for background status changes plus notifications (Ch 50), and periodic/on-open sync (Ch 32) as the reliable backstop so any dropped push or live event self-heals — with the server as truth (the client never invents status), the notification and screen sharing one SSOT, and the whole thing surviving process death and consistent across devices (Ch 8/3/32).**

78.4 High-Level Architecture


78.4 High-Level Architecture

Components & ownership

- **Server** owns the **status state machine** + driver location + ETA (authoritative).
- **Mechanism per app state: live channel** (watching, Ch 63/66 + driver Ch 67) + **FCM** (background + notification, Ch 49/50) + **sync** (backstop, Ch 32) — all into the **order SSOT** (Ch 3).
- **UI + notification share one SSOT**; **lifecycle** closes the live channel off-screen.

Direction: server status → (live channel watching / FCM background / sync backstop) → order SSOT → timeline + map + ETA + consistent notification; self-healing + death-safe.

78.5 Detailed Component Design


78.5 Detailed Component Design

Component	Responsibility
`OrderStatus`	server-owned state machine + timeline + ETA, reflected in SSOT (Ch 3)
`OrderTrackingRepository`	observe SSOT; start/stop live channel (watching); sync backstop
`DriverLocationStream`	watching-scoped smoothed driver location (Ch 67)
`OrderPushHandler`	FCM status change → sync + notification (Ch 49/50)

78.6 End-to-End Data Flow

Status fresh across app states


Status fresh across app states

Self-healing missed update


Self-healing missed update

78.7 Complete Kotlin Implementation

Package: `com.streamcart.ordertracking`.

78.7.1 Server-owned status reflected in the SSOT (Ch 3)

```
@Entity(tableName = "orders")
data class Order(
    @PrimaryKey val id: String,
    val status: OrderStatus,            // server-owned; client NEVER advances it
    val timelineJson: String,           // status events + timestamps (server history)
    val etaMillis: Long?, val driverId: String?, val updatedAt: Long,
)
enum class OrderStatus { Placed, Confirmed, Packed, Shipped, OutForDelivery, Delivered,
Cancelled }
```

78.7.2 Repository: observe SSOT + watching-scoped live channel + sync backstop

```
@Singleton
class OrderTrackingRepository @Inject constructor(
    private val dao: OrderDao, private val api: OrderApi,             // sync (Ch 32)
    private val live: LiveChannel,                                    // WebSocket/SSE (Ch
63/66)
    private val scope: CoroutineScope,
) {
    fun observe(orderId: String): Flow<Order> = dao.observe(orderId)  // SSOT (Ch 3) -> UI

    private var watchJob: Job? = null
    /** While the user WATCHES: low-latency live channel (Ch 63/66) + driver location (Ch 67).
*/
    fun startWatching(orderId: String) {
        watchJob = scope.launch {
            live.subscribeOrder(orderId).collect { update -> dao.applyServerStatus(orderId,
update) }  // server-owned; reflect only
        }
    }
    fun stopWatching() { watchJob?.cancel(); live.unsubscribe() }     // close on leave
(battery, Ch 44/47)

    /** Backstop: pull authoritative status -> self-heal missed push/live (Ch 32). */
    suspend fun sync(orderId: String) = dao.applyServerStatus(orderId,
api.order(orderId).toUpdate())
}
```

78.7.3 FCM status push → sync + notification (Ch 49/50)

```
// High-priority FCM delivers background status changes; trigger a sync + post a notification
(Ch 49/50).
class OrderPushHandler @Inject constructor(
    private val repo: OrderTrackingRepository, private val notifier: Notifier,   // Ch 50
) {
    suspend fun onStatusPush(orderId: String) {
        repo.sync(orderId)                                          // pull authoritative status
(self-heal, Ch 32)
        val order = repo.observe(orderId).first()
        notifier.notifyFor(orderStatusNotification(order))         // notification matches in-
app status (one SSOT)
    }
    // Tapping the notification deep-links (Ch 52) to the tracking screen showing the SAME
status.
}
```

78.7.4 Live driver location (watching-scoped, smoothed, Ch 67)

```
class DriverLocationStream @Inject constructor(private val live: LiveChannel) {
    /** Driver location streams ONLY while watching (Ch 44/67); smoothed; paired with server
ETA. */
    fun observe(orderId: String): Flow<DriverPosition> = live.subscribeDriver(orderId)
        .map { it.smoothed() }                                      // filter/smooth (Ch 67)
    // Closing the tracking screen closes this stream (battery). Driver side = location FGS (Ch
67).
}
```

78.7.5 UI: timeline + map + ETA (one SSOT), lifecycle-scoped live channel (Ch 44)

```
@Composable
fun OrderTrackingScreen(orderId: String, vm: OrderTrackingViewModel = hiltViewModel()) {
    val order by vm.order(orderId).collectAsStateWithLifecycle(initialValue = null)   // SSOT (Ch 3)
    val lifecycleOwner = LocalLifecycleOwner.current
    DisposableEffect(lifecycleOwner) {                                                // live channel only while watching
        vm.startWatching(orderId); vm.syncOnOpen(orderId)                             // + sync backstop (Ch 32)
        onDispose { vm.stopWatching() }                                               // close on leave (Ch 44/47)
    }
    order?.let { o ->
        StatusTimeline(o.timeline, current = o.status)   // server-owned timeline; client renders, never invents
        if (o.status == OrderStatus.OutForDelivery) DriverMap(vm.driverLocation(orderId), eta = o.etaMillis)  // Ch 67
    }
    // Process death: SSOT (Ch 3/8) restores the screen; sync reconciles (Ch 32).
}
```

78.7.6 Module structure

```
:feature:ordertracking  Order/OrderStatus/Dao (SSOT, Ch 3), OrderTrackingRepository (live + sync), DriverLocationStream, OrderPushHandler
:core:realtime  live channel (WebSocket/SSE, Ch 63/66); :feature:push FCM (Ch 49); :core:sync (Ch 32); :core:notifications (Ch 50)
# Server owns status state machine + driver location + ETA. Checkout -> Ch 75; live location -> Ch 67.
```

78.8 Concurrency and Race Conditions

Scenario	Risk	Handling
Client advances status	wrong/invalid state	server-owned; client reflects only (Ch 3)
Dropped push/live event	stale status	sync backstop self-heals (Ch 32)
Out-of-order status events	regress status	apply by server timestamp/sequence; server authoritative
Live channel held in background	battery	close on leave; FCM for background (Ch 44/47/49)
Notification vs screen mismatch	confusion	one SSOT drives both (Ch 3/50)
Process death on tracking screen	lost state	SSOT + sync restore (Ch 3/8/32)
Driver stream off-screen	battery	watching-scoped (Ch 67)

- **Server-owned status + sync backstop** make tracking correct and self-healing: the client **never invents** status (invalid/regressed states), applies updates **by server timestamp/sequence**, and any missed push/live event is **reconciled by sync** (Ch 32) — the SSOT always converges to server truth.
- **Mechanism-per-state** keeps it battery-sane and fresh: the **live channel/driver stream run only while watching** (Ch 44/47), **FCM** handles background, **sync** backstops.
- **One SSOT** guarantees the **notification and the screen agree** (Ch 3/50).

78.9 Failure Scenarios and Recovery

Failure	Detection	Response	Outcome
Status stale/missed	gap	sync backstop (Ch 32)	self-heals
Client-invented status	wrong UI	reflect server-owned only (Ch 3)	correct
Battery drain	live channel in bg	close on leave; FCM background (Ch 44/49)	battery ok
Notification mismatches screen	confusion	one SSOT drives both (Ch 3/50)	consistent
Process death	lost screen	SSOT + sync restore (Ch 8/32)	restored
Driver map laggy	not live while watching	live channel + smoothing while watching (Ch 63/67)	smooth
Background status missed	user unaware	FCM + notification (Ch 49/50)	delivered
Device divergence	user report	server-authoritative + sync (Ch 32)	converge



> **Order tracking is mechanism-selection over a server-owned status machine: live channel while watching, FCM when backgrounded, sync as the backstop — all into one SSOT.** The client reflects the server's status (never invents it), the live channel/driver stream run only while watching (battery), and sync guarantees no update is ever lost.

78.10 Scaling the Implementation

- **Level 1.** Server-owned status in the SSOT (Ch 3); sync on open (Ch 32); status timeline.
- **Level 2.** FCM status changes + notifications (Ch 49/50); live channel (Ch 63/66) while watching; sync backstop self-heals.
- **Level 3 — production.** Live driver location + ETA while watching (Ch 67); lifecycle- scoped channel (Ch 44); notification/screen consistency; process-death survival (Ch 8).
- **Level 4 — large.** Multi-order tracking; ETA refinement; map clustering; proactive alerts (delays); cross-device consistency (Ch 32).
- **Level 5 — enterprise.** Tracking SLOs (freshness/latency), delivery analytics, carrier integrations, real-time fleet scale, proactive issue detection.

Avoid overengineering: server-owned status + FCM + sync + a watching-scoped live channel is the core. Don't poll; don't invent status; don't hold the live channel in the background.

78.11 Alternative Approaches and Trade-offs

Dimension	Option A	Option B	When
Status source	Server-owned (client reflects)	client-inferred	Server always (authoritative state machine)
Freshness	Mechanism per state (live/FCM/sync)	polling	Per-state (fresh + battery-sane)
Live updates	Channel only while watching (Ch 63/66)	always-on socket	While-watching (battery, Ch 44/47)
Background	FCM (Ch 49)	keep socket open	FCM (battery/Doze)
Reliability	Sync backstop (Ch 32)	trust push/live	Sync backstop (self-heal)
Driver map	Live location while watching (Ch 67)	poll location	Live + smoothing while watching
Notification	One SSOT (Ch 3/50)	separate status	One SSOT (consistent)

Server-owned status vs client-inferred: an order's status is a **server-owned state machine** (valid transitions, authoritative timestamps) — the client **cannot** correctly infer or advance it (it lacks fulfillment/logistics data). The client **reflects** the server's status in the SSOT (Ch 3) and **renders** it; inventing status produces wrong or regressed states.

Mechanism per state vs polling vs always-on socket: polling is laggy/wasteful; an **always-on WebSocket** drains battery and is killed in the background (Ch 47). The right design uses the **mechanism per app state** (as in Ch 63/66): a **live channel while watching** (low-latency + driver map), **FCM for background** (+ notifications), and **sync as the backstop**. This maximizes freshness while respecting battery.

Sync backstop vs trusting push/live: FCM and live channels are **best-effort** — a dropped update would leave the status stale forever if trusted blindly. **Sync** (Ch 32) on open/periodic reconciles the SSOT against the authoritative server status, so every missed update **self-heals**. Push/live accelerate; **sync guarantees**.

78.12 Security & Privacy Considerations

- **Server-authorized order access (Ch 23/19):** a user can only track **their own** orders; the server authorizes access to status, timeline, and **driver location** — never a public endpoint. The live channel/FCM topic for an order is **authorized** (Ch 63/49).
- **Driver location privacy (Ch 67/28):** the driver's live location is **sensitive** — shared only **during active delivery**, only with the **authorized customer**, and **stops** on delivery; coarsen/limit as policy requires; don't log coordinates.
- **Notification content (Ch 50/28):** status notifications shouldn't leak sensitive details on the lock screen (public/private versions); don't include PII.
- **Per-account isolation (Ch 42):** orders/tracking scoped to the authenticated user; clear on logout.
- **Untrusted payloads (Ch 49/63/23):** push/live status messages are **untrusted triggers** — validate and reconcile via authed sync (the server status is truth), don't act on a payload's claimed status blindly.
- Client vs backend: the **server** owns status, authorizes access, computes ETA, and shares driver location under authorization; the **client** reflects, subscribes (authorized), and renders — it never owns status or exposes location beyond the authorized customer.

78.13 Performance and Resource Optimization

- **Live channel only while watching** (Ch 44/47) — the top battery lever; FCM for background.
- **Sync on open + sparingly periodic** (Ch 32) — not continuous polling.
- **Smooth + sample driver location** (Ch 67) — don't render every tick; coalesce (Ch 63).
- **Cache status/timeline** (SSOT, Ch 3) for instant render + offline view; refresh via sync.
- **FCM for background** avoids a held connection (Ch 49/47).
- **Efficient map rendering** — update the marker, don't rebuild the map.
- **Profiling:** status freshness/latency, live-channel uptime (while watching), battery, sync frequency, notification delivery, ETA accuracy.

78.14 Testing Strategy

Status reflection, mechanism-per-state, and self-healing are testable with fakes.

```
class OrderTrackingTest {
    @Test fun `reflects server status - never advances locally`() = runTest {
        repo.sync("o1")   // server says Shipped
        assertEquals(OrderStatus.Shipped, dao.observe("o1").first().status)
        // no client API advances status
    }
    @Test fun `sync self-heals a missed update`() = runTest {
        // live/FCM dropped Shipped; open -> sync -> SSOT = Shipped (Ch 32)
        repo.sync("o1"); assertEquals(OrderStatus.Shipped, dao.observe("o1").first().status)
    }
    @Test fun `live channel only while watching`() = runTest {
        repo.startWatching("o1"); assertTrue(live.subscribed)
        repo.stopWatching(); assertFalse(live.subscribed)   // closed on leave (Ch 44)
    }
    @Test fun `out-of-order status does not regress`() = runTest { /* apply by server timestamp
*/ }
}

class OrderPushTest {
    @Test fun `FCM status push syncs then notifies consistently`() = runTest {
        pushHandler.onStatusPush("o1")
        assertTrue(notifier.shown.any { it.orderId == "o1" })   // notification matches SSOT
status (Ch 3/50)
    }
}
```

What to test: status reflected from server (never client-advanced); sync self-heals missed updates (Ch 32); live channel start/stop with watching (Ch 44); out-of-order non-regression; FCM push → sync + consistent notification (Ch 49/50); driver stream watching-scoped (Ch 67); process-death restore (Ch 8). Fake live channel/FCM/sync.

78.15 Production Observability

- **Metrics:** status **freshness/latency** (server change → client reflects) by mechanism; live-channel uptime (while watching) + battery; sync frequency + self-heal rate (how often sync corrects a missed update); FCM status delivery + notification open; ETA accuracy; driver-location smoothness.
- **Logs:** `ordertracking{orderId, status, mechanism, synced}` — no coordinates/PII (Ch 28).
- **Debugging:** stale status → sync backstop not running / mechanism gap (Ch 32); invented status → client advancing (reflect server); battery → live channel in background (Ch 44/ 49); notification mismatch → not one-SSOT (Ch 3/50); laggy map → not live while watching (Ch 63/67); missed background update → FCM (Ch 49).
- **Alerting:** status-freshness regression; live-channel battery spike; FCM delivery drop.
- **Dashboards:** status freshness by mechanism; self-heal rate; ETA accuracy.

78.16 System Design Interview Preparation

> ≥25 questions.

Fundamentals

1. *Who owns order status?* The server (state machine) — the client reflects it (Ch 3).
2. *Client invent status?* No — renders server-owned status.
3. *Freshness mechanism?* Per app state: live channel (watching) + FCM (background) + sync (backstop).
4. *Why not poll?* Laggy, wasteful.
5. *Source of status truth?* Server; SSOT reflects it.

Mechanism per state

6. *While watching?* Live channel (WebSocket Ch 63 / SSE Ch 66) — low-latency + driver map.
7. *Backgrounded?* FCM (Ch 49) + notification (Ch 50).
8. *Backstop?* Sync (Ch 32) on open/periodic.
9. *Why close live channel on leave?* Battery/Doze (Ch 44/47).
10. *WebSocket vs SSE here?* SSE fine for one-way status; WebSocket if bidirectional.

Reliability

11. *Dropped push/live event?* Sync self-heals (Ch 32).
12. *Out-of-order?* Apply by server timestamp/sequence (no regress).
13. *Process death?* SSOT + sync restore (Ch 8/32).
14. *Cross-device?* Server-authoritative + sync (Ch 32).

Notifications

15. *Status notifications?* FCM-driven (Ch 49/50), consistent with in-app (one SSOT).
16. *Tap behavior?* Deep-link to tracking screen (Ch 52), same status.

Live location

17. *Driver map?* Live location (Ch 67) while watching; smoothed; server ETA.
18. *When streamed?* Only during active delivery while watching.
19. *Driver side?* `location` FGS (Ch 67).

Security

20. *Access?* Server-authorized (own orders only, Ch 23).
21. *Driver location privacy?* Sensitive — authorized customer only, during delivery (Ch 67/28).
22. *Payloads?* Untrusted triggers; reconcile via authed sync (Ch 49/63/23).

Testing/Production

23. *Test self-heal?* Dropped update → sync corrects (Ch 32).
24. *Test lifecycle?* Live channel start/stop with watching.
25. *Stale status in prod?* Sync backstop/mechanism gap.

26. *Key metric?* Status freshness/latency + self-heal rate.

Design-change challenges

1. **Live order status without polling.** Server-owned status + FCM + live channel (watching)
    - sync backstop.
2. **Driver on a map.** Live location (Ch 67) while watching; smoothed; server ETA.
3. **Status missed while app closed.** FCM + notification; sync self-heals on reopen (Ch 49/32).
4. **Battery drain from tracking.** Live channel only while watching; FCM background (Ch 44/ 49).
5. **Notification says shipped but screen says packed.** One SSOT drives both (Ch 3/50).

Mock walkthrough (abridged)

Clarify (status timeline, live map, background?) → server-owned status state machine in SSOT (Ch 3) → mechanism per app state: live channel while watching (Ch 63/66 + driver Ch 67) + FCM background + notification (Ch 49/50) + sync backstop (Ch 32) → self-healing + process death (Ch 8) → notification/screen one SSOT → lifecycle closes live channel (Ch 44/47) → security (authorized access, driver-location privacy) → observability (freshness, self-heal rate).

78.17 Practical Exercises

1. **Beginner.** Reflect a server-owned order status in the SSOT and render a timeline; sync on open. *Eval:* client never advances status; timeline correct.
2. **Intermediate.** Add FCM status changes + notifications (consistent with in-app) and a sync backstop that self-heals missed updates. *Eval:* background updates + notification; dropped update heals.
3. **Advanced.** Open a live channel (Ch 63/66) + smoothed driver location (Ch 67) only while watching; close on leave. *Eval:* low-latency map while watching; no background channel.
4. **Production debugging.** Fix stale status (no sync backstop) and battery drain (live channel held in background). *Eval:* fresh via sync; channel scoped to watching.
5. **System design challenge.** Ensure authorized order access + driver-location privacy and a status-freshness dashboard. *Eval:* own-orders-only; location authorized/time-boxed; freshness measured.

(Solutions withheld unless requested.)

78.18 Chapter Summary & Reference Sheet

Key decisions: model order status as a **server-owned state machine** (valid transitions

- authoritative timestamps) **reflected in the SSOT** (Ch 3) — the client **renders, never invents** status; keep it fresh with the **right mechanism per app state** — a **live channel** (WebSocket Ch 63 / SSE Ch 66) **only while the user is actively watching** (low-latency status + the **smoothed driver location and server ETA**, Ch 67, closed on leave for battery, Ch 44/47), **high-priority FCM** (Ch 49) for **background** status changes plus **notifications** (Ch 50), and **periodic/on-open sync** (Ch 32) as the **reliable backstop** so any dropped push or live event **self-heals**; drive **notifications from the same SSOT** so the notification and screen always agree

(and deep-link to tracking, Ch 52); **survive process death** and stay **consistent across devices** (Ch 8/3/32); and **authorize** order access and **protect driver-location privacy** (own orders only, delivery-time-boxed, Ch 23/67/28). Push/live are **accelerators**; **sync is the guarantee**; the **server is truth**.

Concepts: server-owned status state machine reflected in SSOT, mechanism per app state (live channel watching + FCM background + sync backstop), self-healing via sync, notification/screen consistency (one SSOT), live driver location + ETA (watching-scoped), process death, cross-device consistency, authorized access + location privacy.

Common mistakes: client-invented/advanced status; polling for status; always-on WebSocket in the background (battery/Doze); trusting push/live without a sync backstop (stale on drop); notification inconsistent with in-app status (two sources); holding the driver stream off-screen; unauthorized order/location access.

Checklist:

- [ ] server-owned status reflected in SSOT (never invent, Ch 3)
- [ ] live channel (Ch 63/66) only while watching + driver location + ETA (Ch 67)
- [ ] FCM background status + notification (Ch 49/50)
- [ ] sync backstop self-heals missed updates (Ch 32)
- [ ] one SSOT drives screen + notification
- [ ] close live channel on leave (Ch 44/47)
- [ ] process-death survival + cross-device (Ch 8/32)
- [ ] authorized order access + driver-location privacy (Ch 23/67/28)
- [ ] untrusted payloads reconciled via authed sync
- [ ] status-freshness + self-heal telemetry.

Key APIs: Order SSOT (Room `@Entity` / `@Dao`, Ch 3), live channel (WebSocket/SSE, Ch 63/66) watching-scoped via `DisposableEffect` /lifecycle (Ch 44), FCM high-priority (Ch 49) + Notifier (Ch 50) + deep links (Ch 52), delta sync backstop (Ch 32), live driver location (Ch 67, smoothed), server-computed ETA, authorized access (Ch 23).

References:

- Offline-first / sync: https://developer.android.com/topic/architecture/data-layer/offline-first
- (composes) Ch 49 FCM, Ch 63 WebSocket, Ch 66 SSE, Ch 32 Sync, Ch 3 SSOT, Ch 67 Live Location, Ch 50 Notifications, Ch 44 Lifecycle, Ch 75 Checkout

Chapter review (Phase 4)

- **Completeness:** all 18 sections; architecture/class/sequence/flow/failure diagrams; server-owned status SSOT + watching-scoped live channel + sync backstop + FCM push handler
  - driver stream + lifecycle-scoped UI code; concurrency (server-owned, self-heal, mechanism-per-state, one-SSOT); failure (mechanism-selection-over-server-status principle); 5 levels; server-vs-inferred, mechanism-per-state, sync-backstop trade-offs; security (authorized access, driver-location privacy, untrusted payloads); perf (live-while-watching, sync sparingly); self-heal + lifecycle + push tests; observability (freshness, self-heal rate); 26 Q&A + 5 challenges. ✅
- **Expand later:** FCM (Ch 49), WebSocket/SSE (Ch 63/66), sync (Ch 32), live location (Ch 67), notifications (Ch 50).
- **Interview coverage:** exceeds ≥25 bar.